

# C3M2\_peer\_reviewed

August 10, 2024

## 1 C3M2: Peer Reviewed Assignment

### 1.0.1 Outline:

The objectives for this assignment:

1. Apply Poisson Regression to real data.
2. Learn and practice working with and interpreting Poisson Regression Models.
3. Understand deviance and how to conduct hypothesis tests with Poisson Regression.
4. Recognize when a model shows signs of overdispersion.

General tips:

1. Read the questions carefully to understand what is being asked.
2. This work will be reviewed by another human, so make sure that you are clear and concise in what your explanations and answers.

```
[18]: # Load the required packages  
library(MASS)
```

## 2 Problem 1: Poisson Estimators

Let  $Y_1, \dots, Y_n \stackrel{i}{\sim} \text{Poisson}(\lambda_i)$ . Show that, if  $\eta_i = \beta_0$ , then the maximum likelihood estimator of  $\lambda_i$  is  $\hat{\lambda}_i = \bar{Y}$ , for all  $i = 1, \dots, n$ .

### 2.1 Answer

$$Y_i \sim \frac{\lambda_i^{Y_i} \exp\{-\lambda_i\}}{Y_i!}$$

$$f(Y_1, \dots, Y_n) = \prod_{i=1}^n \frac{\lambda_i^{Y_i} \exp\{-\lambda_i\}}{Y_i!}$$

$$\log(f(Y_1, \dots, Y_n)) = \sum_{i=1}^n \{Y_i \times \log(\lambda_i) - \lambda_i - \log(Y_i!)\}$$

Since  $\lambda_i = \exp\{\beta_0\} \forall i \in \{1, 2, \dots, n\}$ . Therefore,  $\lambda_i = \lambda \forall i \in 1, 2, \dots, n$

$$\frac{\partial f}{\partial \lambda} = \sum_{i=1}^n \left(\frac{Y_i}{\lambda} - 1\right)$$

Equating the above expression to 0, we get:

$$\frac{\sum_{i=1}^n Y_i}{\lambda} - n = 0$$

$$\hat{\lambda} = \frac{\sum_{i=1}^n Y_i}{n}$$

\$ = Y} \$

## 2.2 \*\*\*\*\*

### 3 Problem 2: Ships data

The ships dataset gives the number of damage incidents and aggregate months of service for different types of ships broken down by year of construction and period of operation.

The code below splits the data into a training set (80% of the data) and a test set (the remaining 20%).

```
[19]: data(ships)
ships = ships[ships$service != 0,]
ships$year = as.factor(ships$year)
ships$period = as.factor(ships$period)

set.seed(11)
n = floor(0.8 * nrow(ships))
index = sample(seq_len(nrow(ships)), size = n)

train = ships[index, ]
test = ships[-index, ]
head(train)
summary(train)
test
```

|                     |      | type   | year    | period   | service   | incidents |
|---------------------|------|--------|---------|----------|-----------|-----------|
|                     |      | <fct>  | <fct>   | <fct>    | <int>     | <int>     |
| A data.frame: 6 × 5 | 40   | E      | 75      | 75       | 542       | 1         |
|                     | 28   | D      | 65      | 75       | 192       | 0         |
|                     | 18   | C      | 60      | 75       | 552       | 1         |
|                     | 19   | C      | 65      | 60       | 781       | 0         |
|                     | 5    | A      | 70      | 60       | 1512      | 6         |
|                     | 32   | D      | 75      | 75       | 2051      | 4         |
| type                | year | period | service |          | incidents |           |
| A:5                 | 60:7 | 60:11  | Min.    | : 45.0   | Min.      | : 0.00    |
| B:5                 | 65:8 | 75:16  | 1st Qu. | : 318.5  | 1st Qu.   | : 0.50    |
| C:6                 | 70:8 |        | Median  | : 1095.0 | Median    | : 2.00    |
| D:7                 | 75:4 |        | Mean    | : 5012.2 | Mean      | :10.63    |
| E:4                 |      |        | 3rd Qu. | : 2202.5 | 3rd Qu.   | :11.50    |
|                     |      |        | Max.    | :44882.0 | Max.      | :58.00    |

|                     |    | type<br><fct> | year<br><fct> | period<br><fct> | service<br><int> | incidents<br><int> |
|---------------------|----|---------------|---------------|-----------------|------------------|--------------------|
| A data.frame: 7 × 5 | 1  | A             | 60            | 60              | 127              | 0                  |
|                     | 4  | A             | 65            | 75              | 1095             | 4                  |
|                     | 10 | B             | 60            | 75              | 17176            | 29                 |
|                     | 16 | B             | 75            | 75              | 7117             | 18                 |
|                     | 21 | C             | 70            | 60              | 783              | 6                  |
|                     | 35 | E             | 65            | 60              | 789              | 7                  |
|                     | 37 | E             | 70            | 60              | 1157             | 5                  |

### 3.0.1 2. (a) Poisson Regression Fitting

Use the training set to develop an appropriate regression model for incidents, using type, period, and year as predictors (HINT: is this a count model or a rate model?).

Calculate the mean squared prediction error (MSPE) for the test set. Display your results.

```
[20]: model_full = glm(incidents~type+period+year, family=poisson("log"), data=train)
summary(model_full)
predictions = exp(predict(model_full, newdata=test))
actual = test$incidents

mse = mean((predictions-actual)^2)
sprintf("The mean square error is: %f", mse)
```

Call:

```
glm(formula = incidents ~ type + period + year, family = poisson("log"),
    data = train)
```

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -4.0775 | -1.9869 | -0.0418 | 0.7612 | 3.6618 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 1.5644   | 0.2199     | 7.113   | 1.13e-12 *** |
| typeB       | 1.6795   | 0.1889     | 8.889   | < 2e-16 ***  |
| typeC       | -2.0789  | 0.4408     | -4.717  | 2.40e-06 *** |
| typeD       | -1.1551  | 0.2930     | -3.943  | 8.06e-05 *** |
| typeE       | -0.5113  | 0.2781     | -1.839  | 0.0660 .     |
| period75    | 0.4123   | 0.1282     | 3.216   | 0.0013 **    |
| year65      | 0.4379   | 0.1885     | 2.324   | 0.0201 *     |
| year70      | 0.2260   | 0.1916     | 1.180   | 0.2382       |
| year75      | 0.1436   | 0.3147     | 0.456   | 0.6481       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

```
Null deviance: 554.70  on 26  degrees of freedom
Residual deviance: 109.21  on 18  degrees of freedom
AIC: 200.92
```

Number of Fisher Scoring iterations: 6

'The mean square error is: 131.077556'

### 3.1 Answer

The mean square error is 131.077556

### 3.2 \*\*\*\*\*

#### 3.2.1 2. (b) Poisson Regression Model Selection

Do we really need all of these predictors? Construct a new regression model leaving out year and calculate the MSPE for this second model.

Decide which model is better. Explain why you chose the model that you did.

```
[21]: model_sub = glm(incidents~type+period, family=poisson("log"), data=train)
summary(model_sub)
predictions = exp(predict(model_sub, newdata=test))
actual = test$incidents

mse = mean((predictions-actual)^2)
sprintf("The mean square error is: %f", mse)
```

Call:

```
glm(formula = incidents ~ type + period, family = poisson("log"),
    data = train)
```

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -4.2377 | -1.9003 | -0.1372 | 0.6377 | 3.8906 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 1.7190   | 0.1838     | 9.355   | < 2e-16 ***  |
| typeB       | 1.7831   | 0.1781     | 10.014  | < 2e-16 ***  |
| typeC       | -2.0573  | 0.4394     | -4.683  | 2.83e-06 *** |
| typeD       | -1.1281  | 0.2918     | -3.866  | 0.000111 *** |
| typeE       | -0.4831  | 0.2767     | -1.746  | 0.080787 .   |
| period75    | 0.4723   | 0.1222     | 3.865   | 0.000111 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 554.70 on 26 degrees of freedom  
Residual deviance: 115.63 on 21 degrees of freedom  
AIC: 201.34

Number of Fisher Scoring iterations: 6

'The mean square error is: 275.122551'

```
[22]: # Can compare nested poisson models with a chi-squared
pchisq(abs(deviance(model_full)-deviance(model_sub)), abs(df.
  ↪residual(model_full)-df.residual(model_sub)), lower.tail = FALSE)
```

0.0929203838345225

### 3.3 Answers

Since the deviances are distributed by chisquare distributions with corresponding degrees of freedoms, their subtraction will also be chisquare with degrees of freedom equal to the difference in the degrees of freedom of the two models' residual deviance.

The chisquare test verdict is that both the models are equivalent with respect to how well they fit to their training data.

However, the smaller model performs well on the test data. Therefore, I am inclined to choose the smaller model (without the year covariate included.)

Note that Mean Sum of Squares is usually not a good measure when comparing nested models because the full model will outperform the model with a subset of covariates, however, that is the case when the models' MSEs are being compared over the training data. Comparison over the test data seems to be valid to this author.

tl;dr - **The smaller model is better** because chisquare test represents the difference in performance over training data and MSE is calculated over the test data.

Chi-square shows no difference, whereas MSE shows some difference in favour of the smaller (without year covariate) model.

### 3.4 \*\*\*\*\*

#### 3.4.1 2. (c) Deviance

How do we determine if our model is explaining anything? With linear regression, we had a F-test, but we can't do that for Poisson Regression. If we want to check if our model is better than the null model, then we're going to have to check directly. In particular, we need to compare the deviances of the models to see if they're significantly different.

Conduct two  $\chi^2$  tests (using the deviance). Let  $\alpha = 0.05$ :

1. Test the adequacy of null model.

2. Test the adequacy of your chosen model against the full model (the model fit to all predictors).

What conclusions should you draw from these tests?

```
[23]: # Your Code Here
# Test if the model is better than the null model
# I am assuming that the model to test here is the chose model, smaller one in
→my case.

# Test chi_sq stat
pchisq(model_sub$null.deviance, model_sub$df.null, lower.tail = FALSE)

# Test against the full model
# I have already done that in the previous cell, so is this a repeat? IDK
pchisq(abs(deviance(model_full)-deviance(model_sub)), abs(df.
→residual(model_full)-df.residual(model_sub)), lower.tail = FALSE)
```

1.5950789625333e-100

0.0929203838345225

### 3.5 Conclusions:

1. The smaller model is significantly different than the null model, i.e., the addition of covariates other than the intercept have significant effect on the performance of the model.
2. Both (small and large/full) models are equivalent with respect to how well they fit to their training data.

### 3.6 \*\*\*\*\*

#### 3.6.1 2. (d) Poisson Regression Visualizations

Just like with linear regression, we can use visualizations to assess the fit and appropriateness of our model. Is it maintaining the assumptions that it should be? Is there a discernable structure that isn't being accounted for? And, again like linear regression, it can be up to the user's interpretation what is an isn't a good model.

Plot the deviance residuals against the linear predictor  $\eta$ . Interpret this plot.

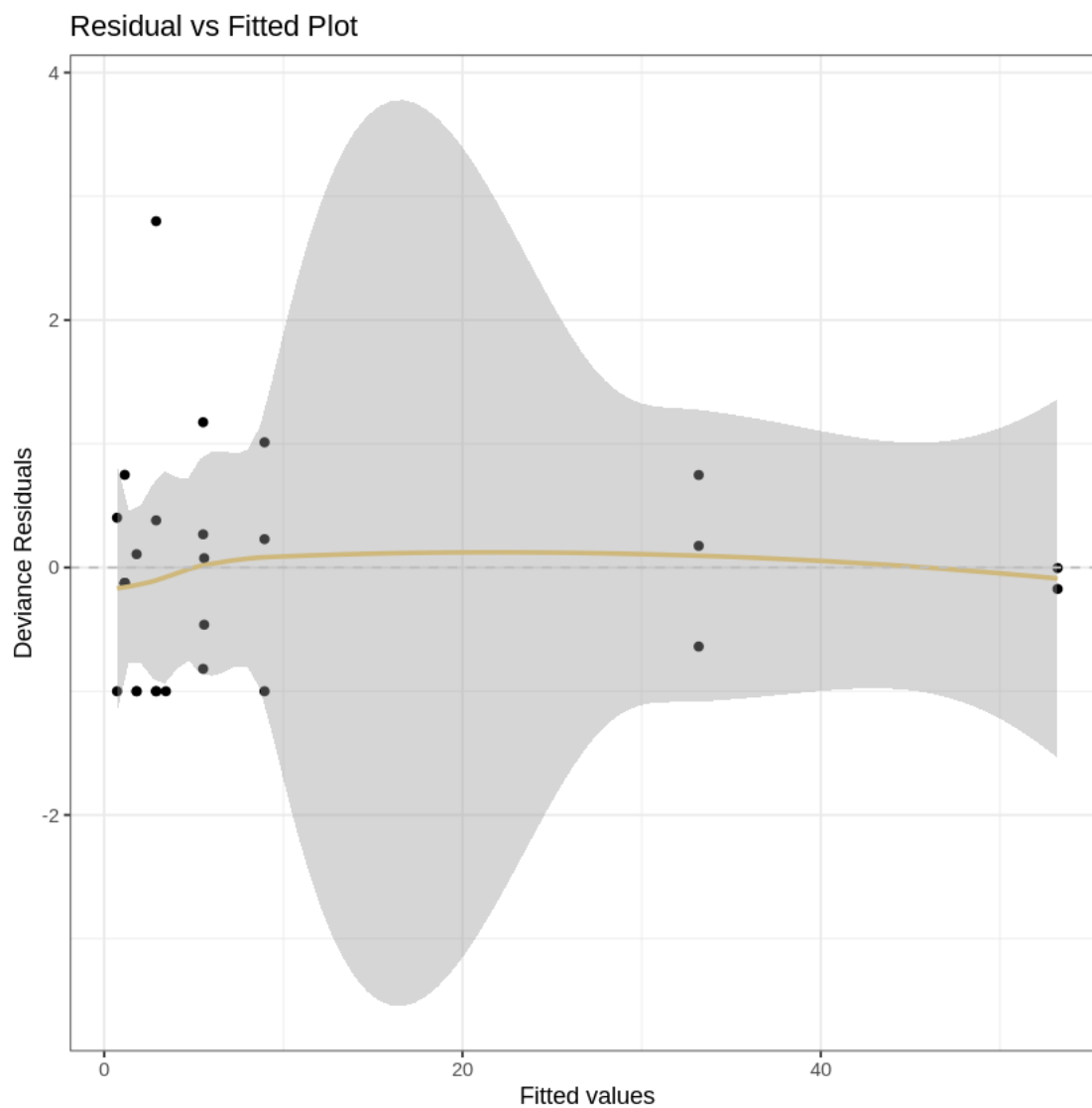
### 3.7 Note for Peer Reviewer

The graph below might look different because the code for creating this graph provided by the instructors in the notbook titled "Poisson regression goodness of fit in R" seems to be inaccurate. I had to make the necessary modifications.

When I tried using the instructor provided code to plot the graph with my model\_sub model, ggplot changed the magnitude of deviance residuals. There were multiple deviance residuals of magnitude greater then 3 in the plot. However when I inspected the actual list of residuals, I could find no such values. There was only one record with magnitude greater than 2.

```
[116]: # Your Code Here
library(tidyverse)
# p1 = ggplot(model_sub, aes(.fitted, .resid))+geom_point()
p1 = ggplot(data=data.frame(x=model_sub$fitted, y=model_sub$residuals), aes(x=x, y=y))+geom_point()
p1 = p1+stat_smooth(method="loess", col = "#CFB87C")+geom_hline(yintercept=0, col="grey", linetype="dashed")
p1 = p1+xlab("Fitted values")+ylab("Deviance Residuals")
p1 = p1+ggtitle("Residual vs Fitted Plot")+theme_bw()
p1
```

`geom\_smooth()` using formula 'y ~ x'



### 3.8 Analysis

Some outlier values do seem to be present in the plot, but overall: 1. the points are randomly scattered about the  $y=0$  line. There is not systemic pattern 2. The variance seems to be equal, however there may be one outlier that may be contributing to overdispersion. Overdispersion may have to be confirmed via the chisq test

### 3.9 \*\*\*\*\*

#### 3.9.1 2. (e) Overdispersion

For linear regression, the variance of the data is controlled through the standard deviation  $\sigma$ , which is independent of the other parameters like the mean  $\mu$ . However, some GLMs do not have this independence, which can lead to a problem called overdispersion. Overdispersion occurs when the observed data's variance is higher than expected, if the model is correct.

For Poisson Regression, we expect that the mean of the data should equal the variance. If overdispersion is present, then the assumptions of the model are not being met and we can not trust its output (or our beloved p-values)!

Explore the two models fit in the beginning of this question for evidence of overdispersion. If you find evidence of overdispersion, you do not need to fix it (but it would be useful for you to know how to). Describe your process and conclusions.

```
[117]: # Your Code Here
dp = sum(residuals(model_sub, type = "pearson")^2)/model_sub$df.res
dp
dp = sum(residuals(model_full, type = "pearson")^2)/model_full$df.res
dp
```

4.93792459262441

5.47049561223492

### 3.10 Analysis

The pearson test statistic divided by the degree of freedom are greater the 1 for both models. Therefore, both of them have over-dispersion.

Some outliers were seen in the deviance residuals vs fitted values graph for model\_sub (sans year). Removing those may reduce the overdispersion.

### 3.11 \*\*\*\*\*

#### 3.12 Fixing Overdispersion

1. remove all records that have deviance residual value  $\geq 2$  according to the model\_sub (see above, it is the model without the year covariate)
2. Add  $\log(\text{service})$  as the offset to the full model (including year covariate)

The reason I had to add  $\log(\text{service})$  as a separate covariate instead on specifying the `offset=service` parameter, was because it the `glm` function gave me the error "Error: no valid set of coefficients has been found: please supply starting values"

I have no idea what this error is and why it is appearing.

### 3.13 The results

1. Pearson statistic to degrees of freedom ratio = 1.07274515153725, which is ~1
2. MSE on test set is 10.167984, which 10 times less than both the prior two nested models' MSEs
3. p-value of the chisquare test of the deviance w.r.t the saturated model is ~0.62, which means it fits well to the training data.

```
[118]: #Removing all the records with
train$deviance_residuals <- model_sub$resid

#### STEP 1
train_sans_outlier <- subset(train[!abs(train['deviance_residuals'])>=2, ],
  ↳select=-c(deviance_residuals))
# train_sans_outlier <- train
# train_sans_outlier$service <- train_sans_outlier$service/12

head(train_sans_outlier)
```

A data.frame: 6 × 5

|    |   | type<br><fct> | year<br><fct> | period<br><fct> | service<br><int> | incidents<br><int> |
|----|---|---------------|---------------|-----------------|------------------|--------------------|
| 40 | E | 75            | 75            | 542             | 1                |                    |
| 28 | D | 65            | 75            | 192             | 0                |                    |
| 18 | C | 60            | 75            | 552             | 1                |                    |
| 19 | C | 65            | 60            | 781             | 0                |                    |
| 5  | A | 70            | 60            | 1512            | 6                |                    |
| 32 | D | 75            | 75            | 2051            | 4                |                    |

```
[119]: #### STEP 2
model_sans_outlier = glm(incidents~type+period+year+log(service),
  ↳family=poisson("log"), data=train_sans_outlier)
```

```
[120]: summary(model_sans_outlier)
```

Call:

```
glm(formula = incidents ~ type + period + year + log(service),
    family = poisson("log"), data = train_sans_outlier)
```

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.3324 | -0.6584 | -0.2689 | 0.4969 | 1.9531 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | -5.8560  | 1.2125     | -4.830  | 1.37e-06 | *** |
| typeB       | -0.4901  | 0.3312     | -1.480  | 0.138944 |     |
| typeC       | -1.3397  | 0.4658     | -2.876  | 0.004022 | **  |

```

typeD      -0.8221      0.4737   -1.735  0.082700 .
typeE      0.1451      0.2916    0.498  0.618738
period75   0.3944      0.1470    2.684  0.007279 **
year65     0.7346      0.2147    3.421  0.000624 ***
year70     0.7660      0.2690    2.848  0.004406 **
year75     0.5204      0.3993    1.303  0.192540
log(service) 0.9353      0.1357    6.891  5.55e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 554.690  on 25  degrees of freedom
Residual deviance: 17.101  on 16  degrees of freedom
AIC: 106.56

```

Number of Fisher Scoring iterations: 5

```

[121]: dp = sum(residuals(model_sans_outlier, type = "pearson")^2)/
      ↪model_sans_outlier$df.res
dp
pchisq(deviance(model_sans_outlier), model_sans_outlier$df.resid)

1.07274515153725
0.620893053122464

```

```

[122]: test_mod <- test
      # test_mod$service <- test$service/12
      predictions = exp(predict(model_sans_outlier, newdata=test_mod))

      actual = test_mod$incidents

      mse = mean((predictions-actual)^2)
      sprintf("The mean square error is: %f", mse)

```

'The mean square error is: 10.167984'

```

[123]: predictions

1      0.2657075036286 4      6.16280853250063 10      23.7712292209917 16      17.5463630366108 21
0.820471252781848 35      3.53495149504675 37      5.21825182480388

```

```

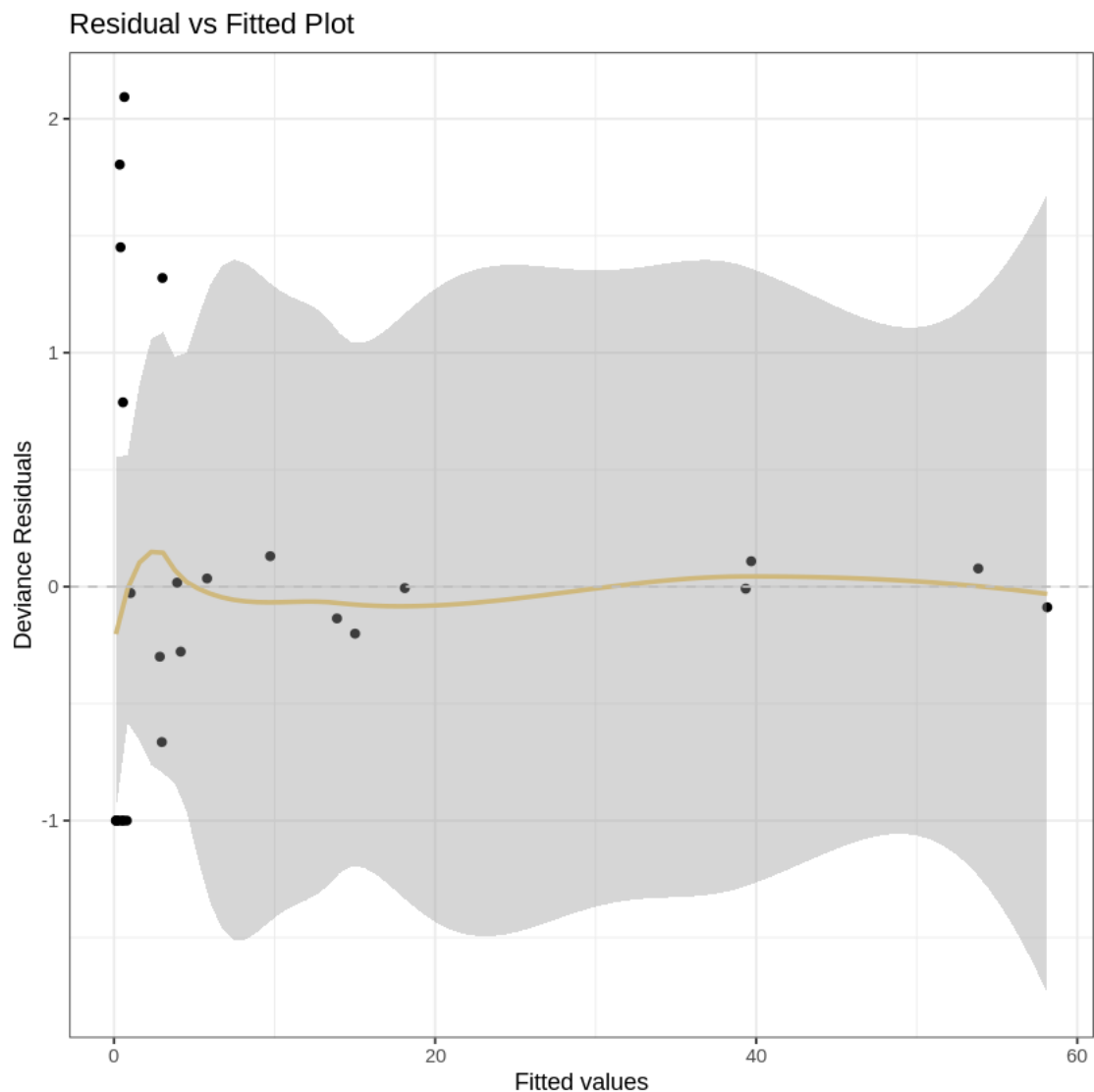
[124]: actual

1. 0 2. 4 3. 29 4. 18 5. 6 6. 7 7. 5

```

```
[127]: library(tidyverse)
# p1 = ggplot(model_sans_outlier, aes(.fitted, .resid))+geom_point()
p1 = ggplot(data=data.frame(x=model_sans_outlier$fitted,
  →y=model_sans_outlier$residuals), aes(x=x, y=y))+geom_point()
p1 = p1+stat_smooth(method="loess", col = "#CFB87C")+geom_hline(yintercept=0,
  →col="grey", linetype="dashed")
p1 = p1+xlab("Fitted values")+ylab("Deviance Residuals")
p1 = p1+ggtitle("Residual vs Fitted Plot")+theme_bw()
p1
```

`geom\_smooth()` using formula 'y ~ x'



[ ]: